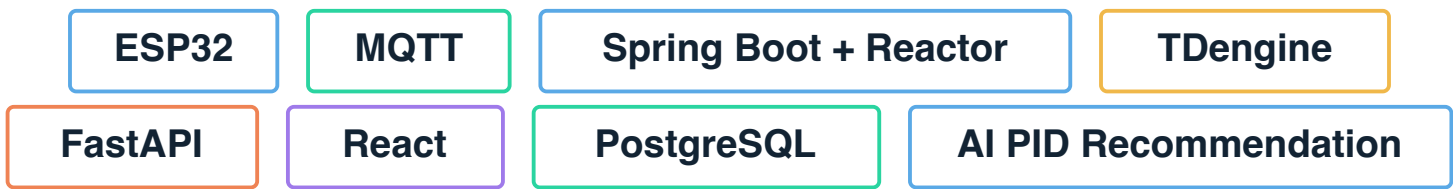
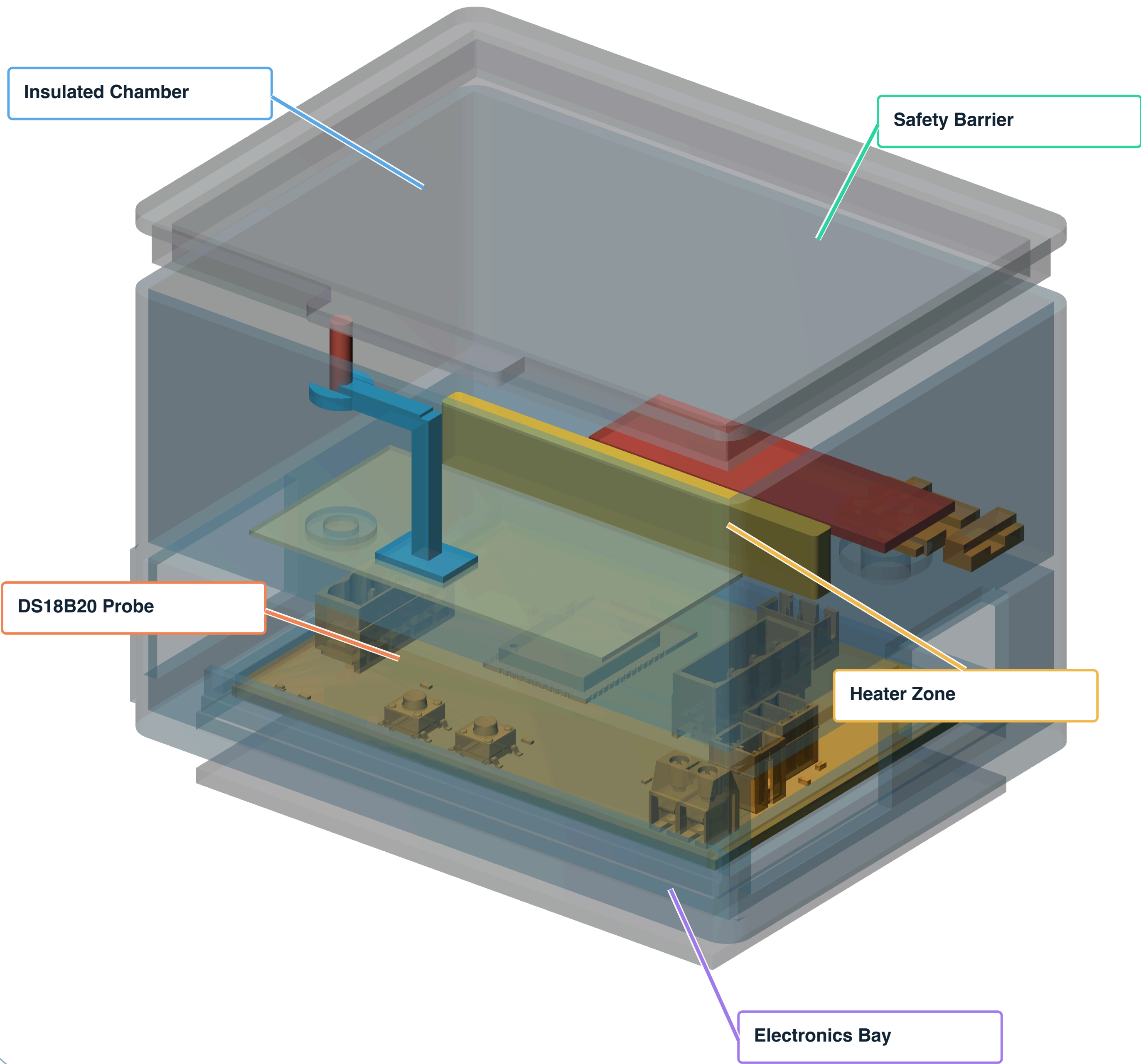


Design of Intelligent Edge and Time-Series Data Flow Temperature Control System



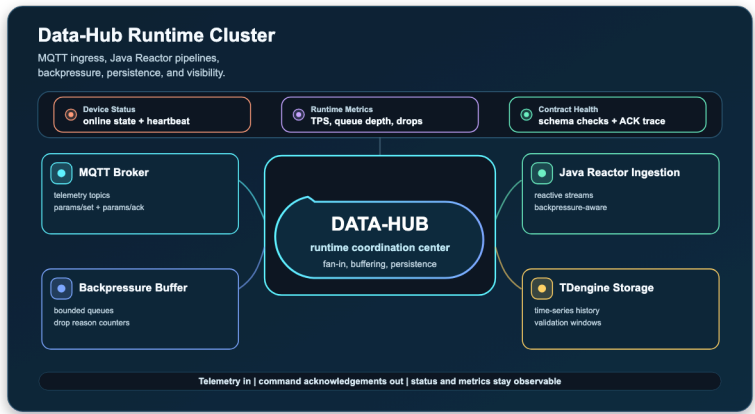
Edge Device / Thermal Enclosure

3D thermal insulation enclosure and edge temperature-control prototype

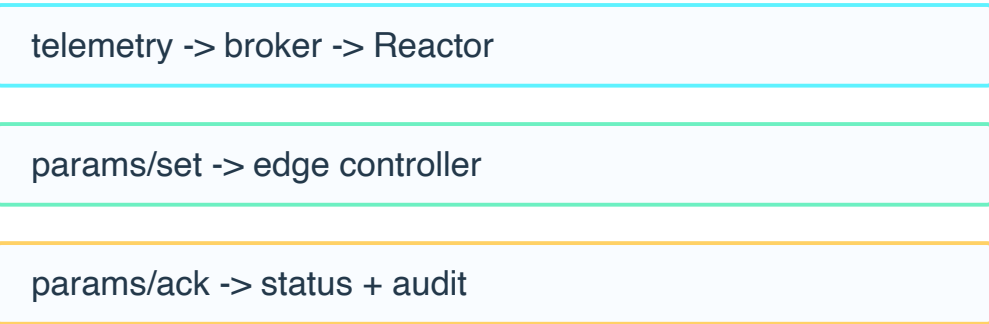


Data-Hub Runtime Cluster

MQTT ingress, Java Reactor ingestion, buffering, persistence, and runtime observability

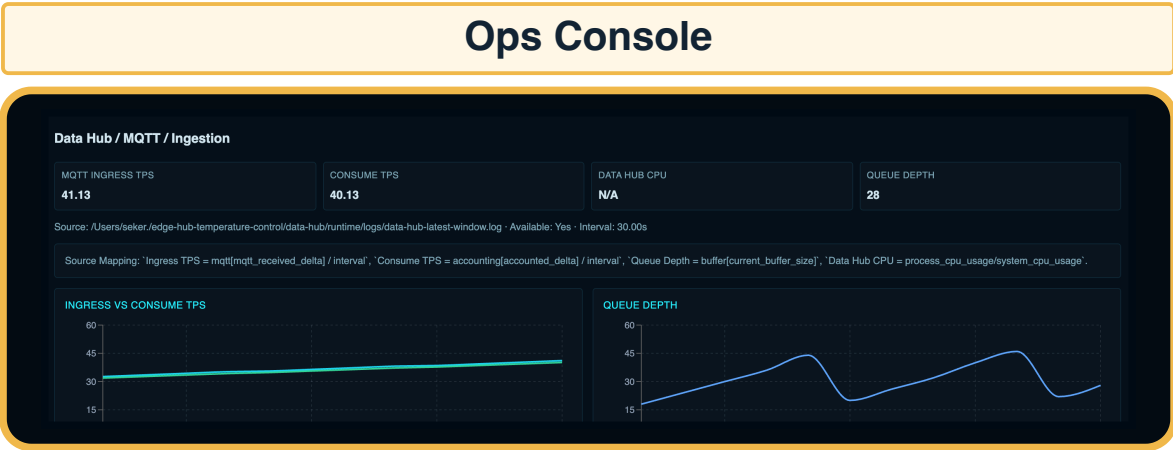
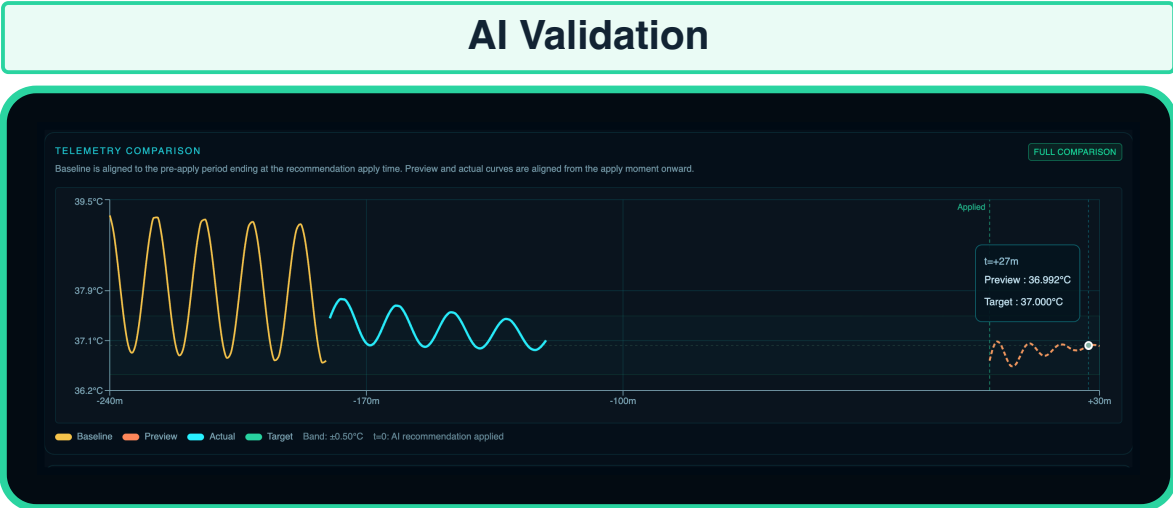
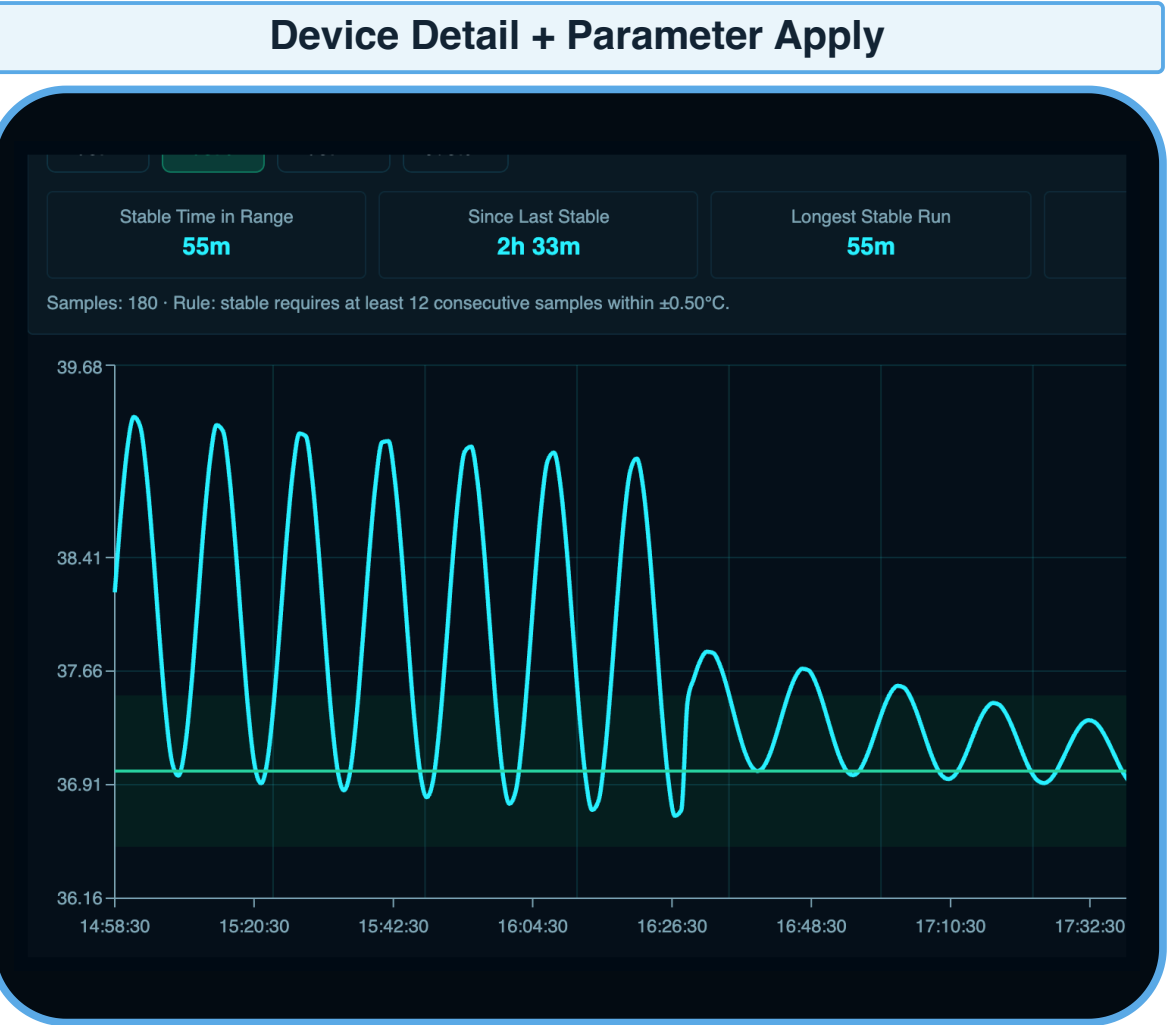


Runtime Contract



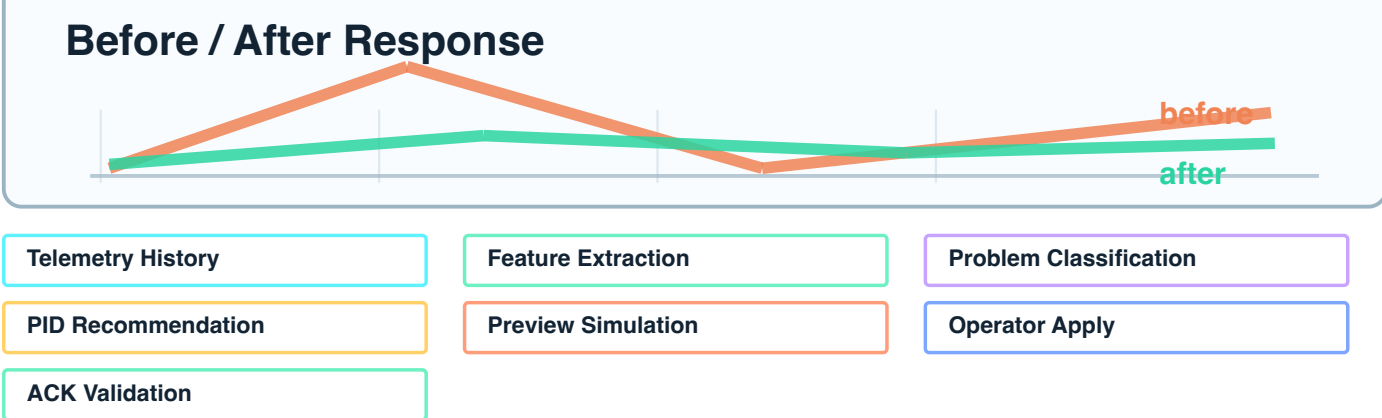
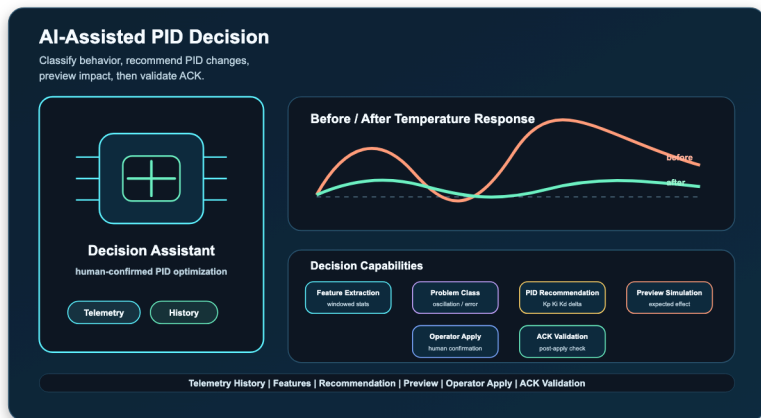
HMI Layer

Real FastAPI + React screenshots with seeded telemetry data



AI-Assisted Decision

features, recommendation, preview, operator apply, and ACK validation



Key Contributions

- 01 Edge closed-loop temperature control
- 02 MQTT runtime telemetry + commands
- 03 Data-Hub time-series persistence
- 04 HMI workflow operate + apply params
- 05 AI PID assist recommend + validate